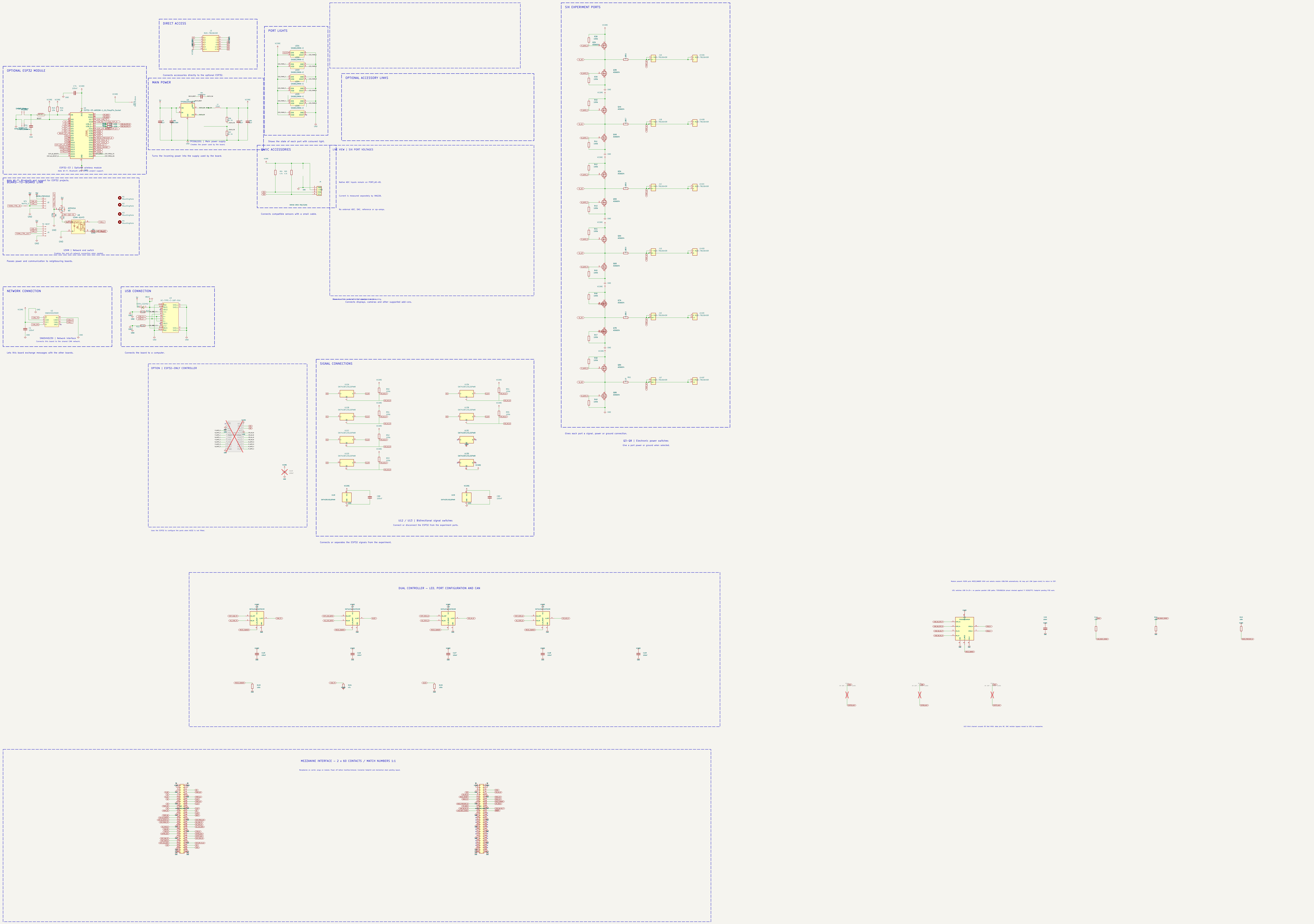
SPAGHETTILAB | ONE BOARD, MANY POSSIBILITIES

Connect a circuit, choose how each port behaves, and explore what happens.

CONNECTIONS & POWER

CARRIER – ESP32 BYPASS, POWER SWITCHING, CAN, USB AND STATUS

No module: MEZZ_OWNER LOW selects ESP USB/CAN; ESP firmware enables U12/U13 for P1–P6. All module measurement paths are absent.



Connects accessories directly to the optional ESP32.

Turns the incoming power into the supply used by the board.

Shows the state of each port with coloured light.

Connects compatible sensors with a small cable.

Native ADC inputs remain on PORT_A0–A5.

Current is measured separately by INA226.

No external ADC, DAC, reference or op-amps.

Connects displays, cameras and other supported add-ons.

Adds Wi-Fi, Bluetooth and support for ESP32 projects.

Passes power and communication to neighbouring boards.

Lets this board exchange messages with the other boards.

Connects the board to a computer.

Gives each port a signal, power or ground connection.

Uses the ESP32 to configure the ports when AG32 is not fitted.

Connects or separates the ESP32 signals from the experiment.

Receptacles on carrier, plugs on module. Power off before insertion/removal. Connector footprint and mechanical stack pending layout.

Module present: R109 pulls MEZZ_OWNER HIGH and selects module USB/CAN automatically. AG may pull LOW (open-drain) to return to ESP.

U31 switches USB D+/D−; no passive parallel USB paths. TS3USB221A pinout checked against TI SCDS277C; footprint pending PCB work.

U13 third channel unused: OE tied HIGH, data pins NC. DAC resistor bypass moved to U32 on mezzanine.

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